

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – INDUSTRY PROBLEM TASK (5 Questions)

Course Code:	DSA0607	Course Title:	Data Handling and Visualization
CO Assessed:	CO2 – Precision (BL3)	Assessment:	Assessment Tool 1 – Industry Problem Task
Weightage:	40% of CIA	Total Marks:	50 (5 Questions × 10 Marks)
CO2 Statement:	<i>Apply appropriate visualization methods to represent distributions, proportions, and relationships in data.(BL3)</i>		
Student Name:	M Jeevitha	Reg. No.:	192424257
Section / Batch:	D/2024	Date:	25.7.2026

Code of Conduct

I M Jeevitha(192424257) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: jeevitha

Instructions

- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Categorical Data Visualization	Comparing category-wise revenue, proportions, and seasonal trends using bar, grouped, and stacked charts	1	10
Distribution Analysis	Understanding defect distribution and variability using histograms, boxplots, and violin plots	2	10
Relationship Visualization	Exploring relationships between variables using scatter plots, heatmaps, and multivariate techniques	3	10
Geospatial & Temporal Visualization	Analyzing delivery performance using maps, time-series plots, and heatmaps	4	10
Distribution Comparison	Comparing patient data distributions using ridgeline plots, density plots, and boxplots	5	10
GRAND TOTAL:		50 Marks	

Annexure A – Industry Problem Task

1.A university wants to analyze the academic performance of students across different departments (Computer Science, Mechanical, Civil, Electronics, etc.). The administration needs to compare average scores, identify top-performing departments, and examine variations in student results.

Task: Design suitable visualizations to represent student scores and performance distributions. Justify the use of bar plots, grouped bars, histograms, and boxplots. Explain how these visualizations help identify performance trends and support academic improvement strategies.

Academic Performance Analysis Using Data Visualization

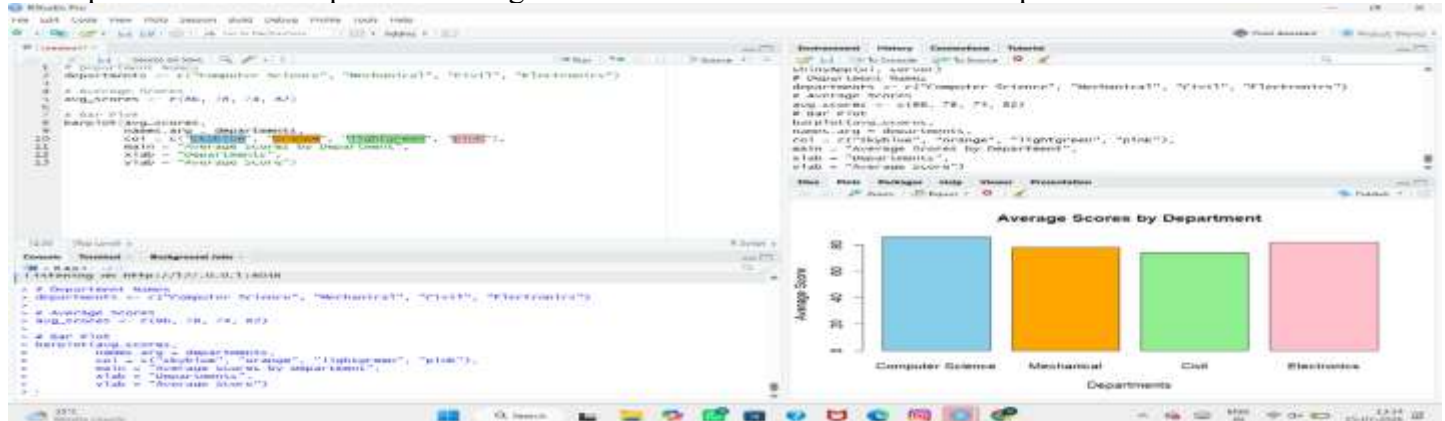
Introduction

Data visualization is an important technique used to analyze and present academic data in a meaningful way. Universities collect large amounts of student performance data from different departments such as Computer Science, Mechanical, Civil, and Electronics. By using suitable visualizations, administrators can compare departmental performance, identify strengths and weaknesses, understand score distributions, and make informed decisions to improve academic quality. Different visualization techniques serve different purposes, making it easier to interpret complex datasets.

1. Bar Plot

Purpose

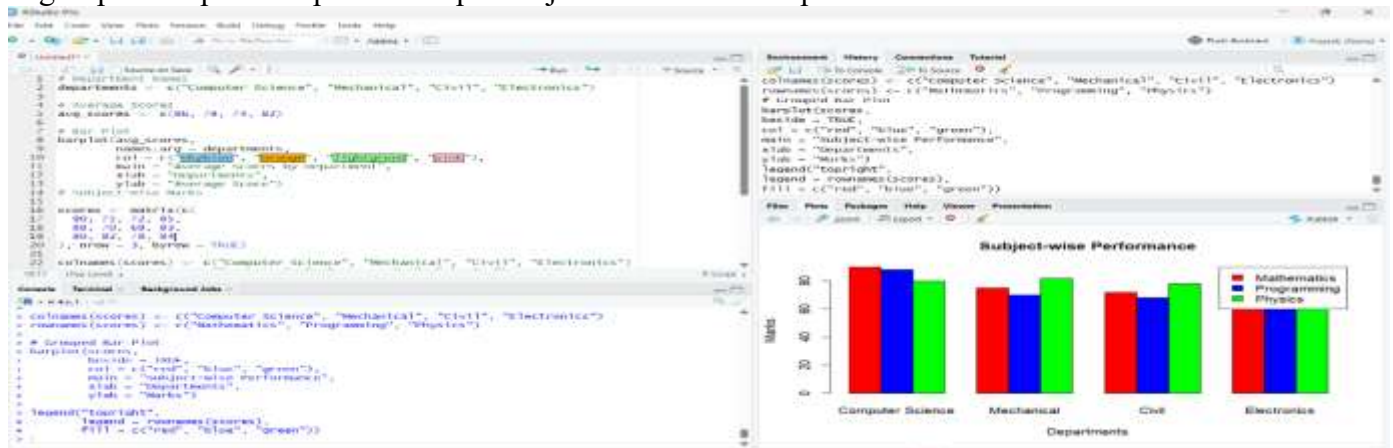
A bar plot is used to compare the average scores of students across different departments.



2. Grouped Bar Plot

Purpose

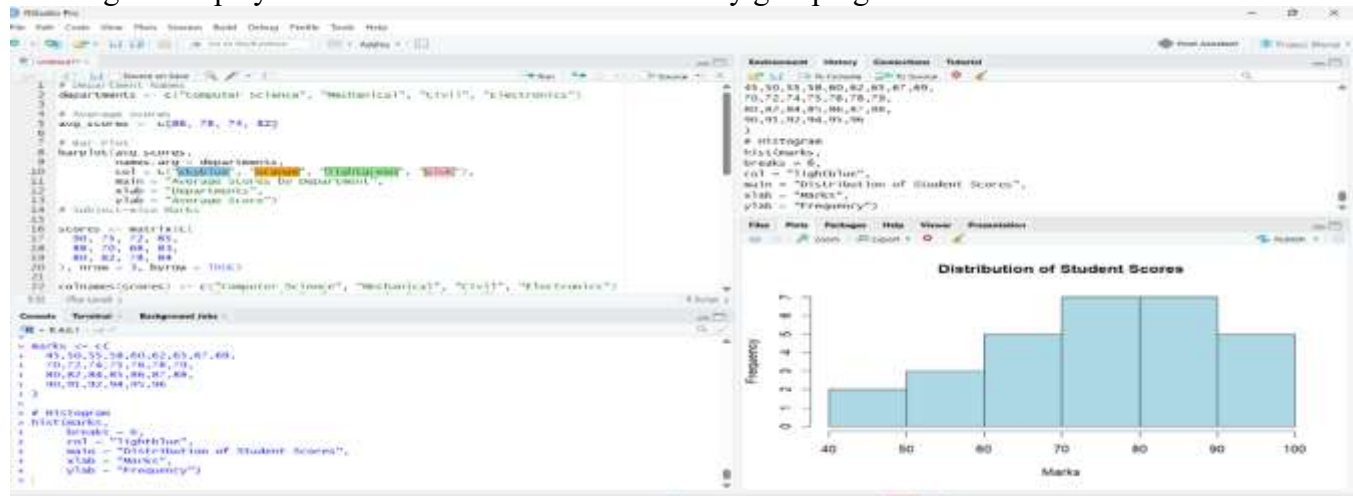
A grouped bar plot compares multiple subjects within each department.



3. Histogram

Purpose:

A histogram displays the distribution of student marks by grouping scores into intervals.



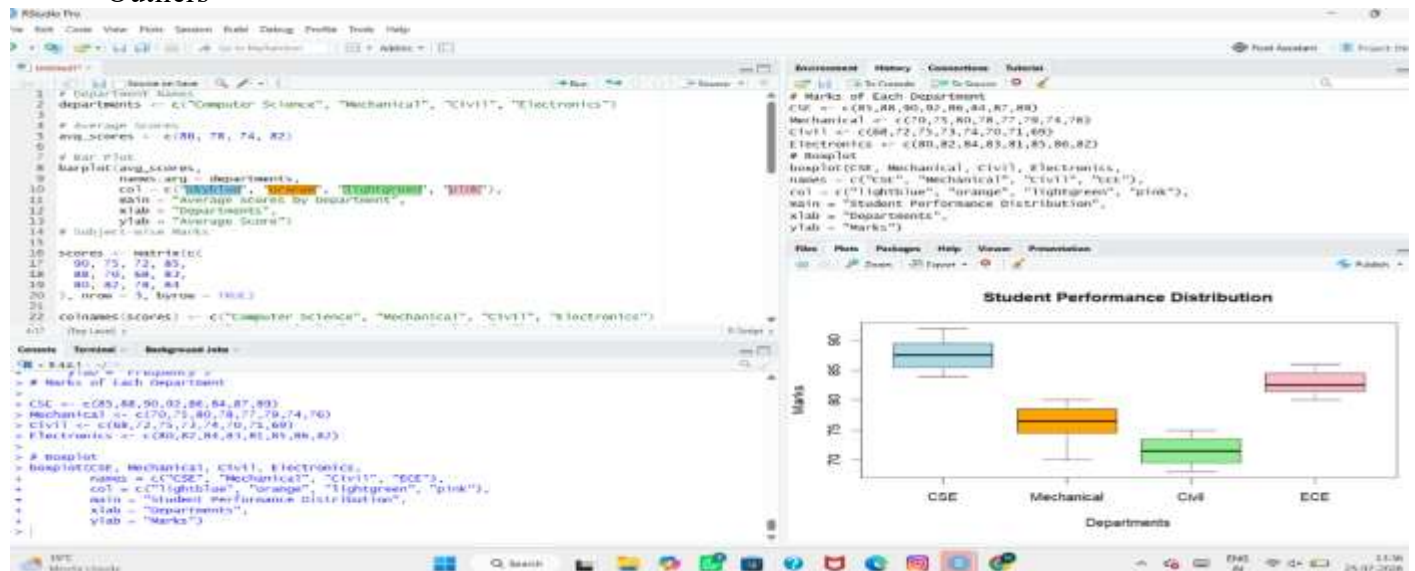
4. Boxplot

Purpose

A boxplot summarizes the distribution of scores using statistical measures.

It displays:

- Minimum score
- First Quartile (Q1)
- Median
- Third Quartile (Q3)
- Maximum score
- Outliers



2.A supermarket chain collects data on customer purchases across various product categories and seasons. The management wants to understand spending patterns, compare sales distributions, and identify categories with the highest contribution to revenue.

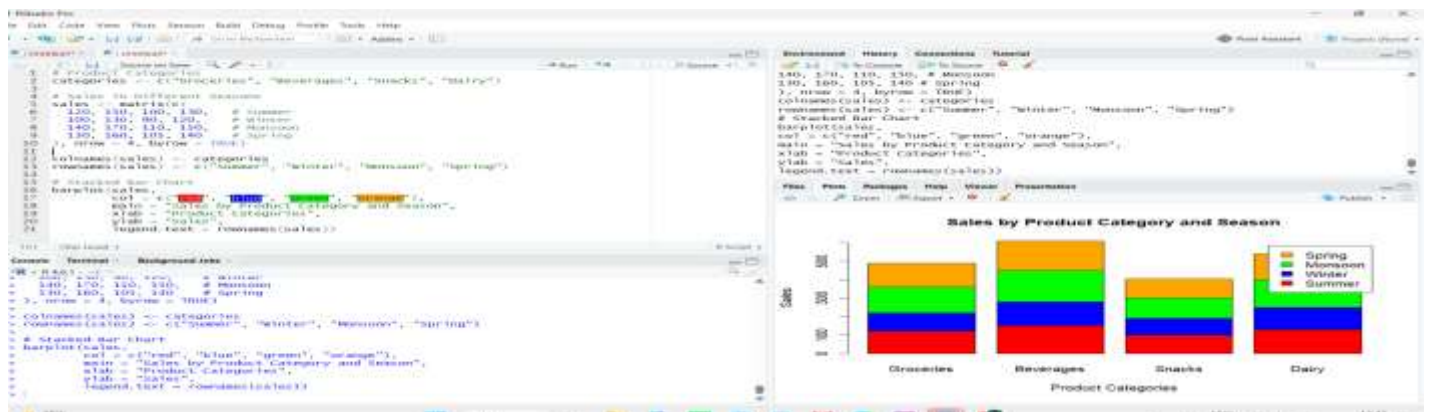
Task: Recommend appropriate visualization techniques such as stacked bars, heatmaps, density plots, and violin plots. Explain how these visualizations help reveal customer preferences, seasonal trends, and business opportunities.

Introduction

A supermarket chain collects customer purchase data across different product categories and seasons to understand buying behavior and improve business decisions. Data visualization helps transform this information into easy-to-understand charts that reveal spending patterns, seasonal demand, sales distribution, and the highest revenue-generating product categories. Using appropriate visualizations enables management to optimize inventory, improve marketing strategies, and increase overall profitability.

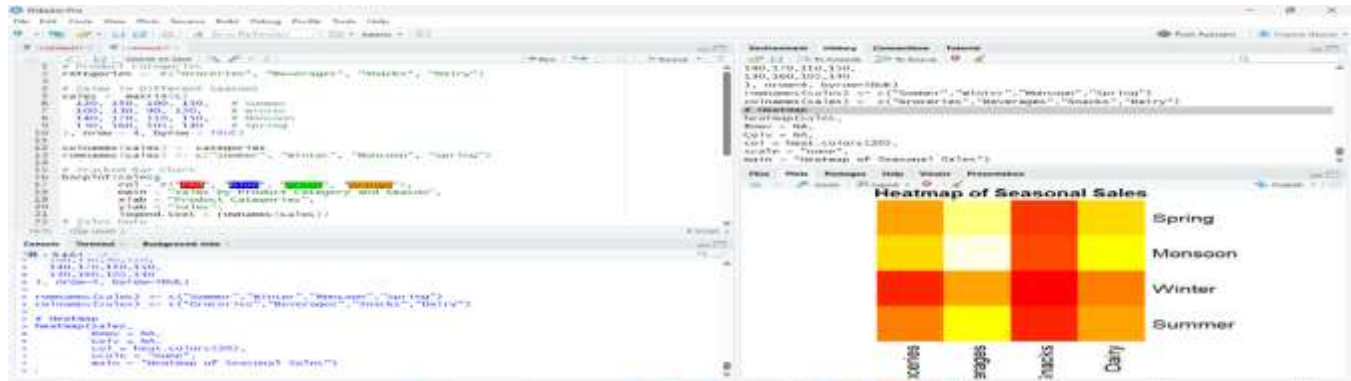
1. Stacked Bar Chart

One-line explanation: A stacked bar chart compares total sales across product categories while showing the contribution of each season to overall revenue.



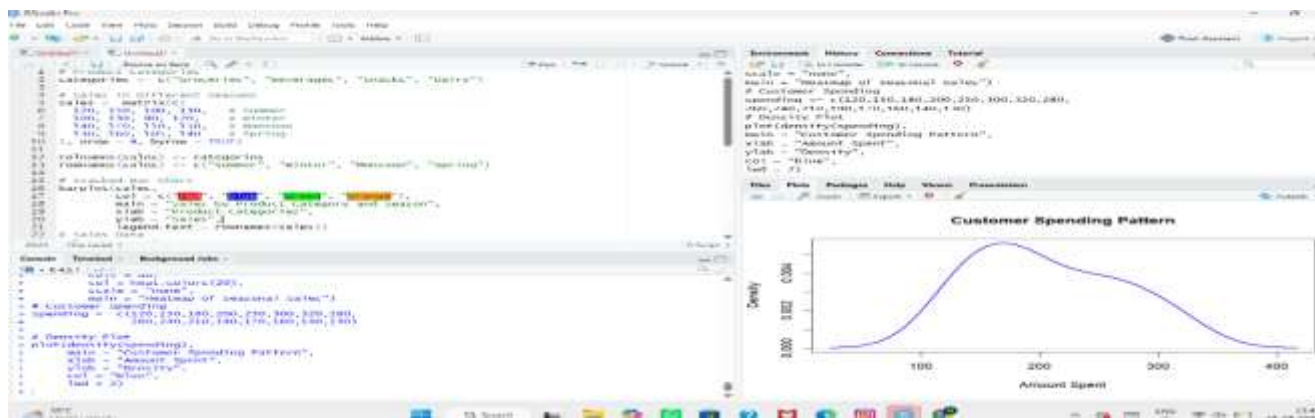
2. Heatmap

One-line explanation: A heatmap uses color intensity to highlight high and low sales, making seasonal trends and popular product categories easy to identify.



3. Density Plot

One-line explanation: A density plot displays the distribution of customer spending, helping identify the most common spending patterns.



4. Violin Plot

- Displays the distribution and spread of sales for each product category.
- Shows median, variability, and concentration of sales.
- Helps compare sales consistency across categories.



3.A smart city project gathers data on vehicle counts, traffic congestion levels, average speeds, and travel times across different locations and time periods. City planners aim to identify congestion hotspots and evaluate transportation efficiency.

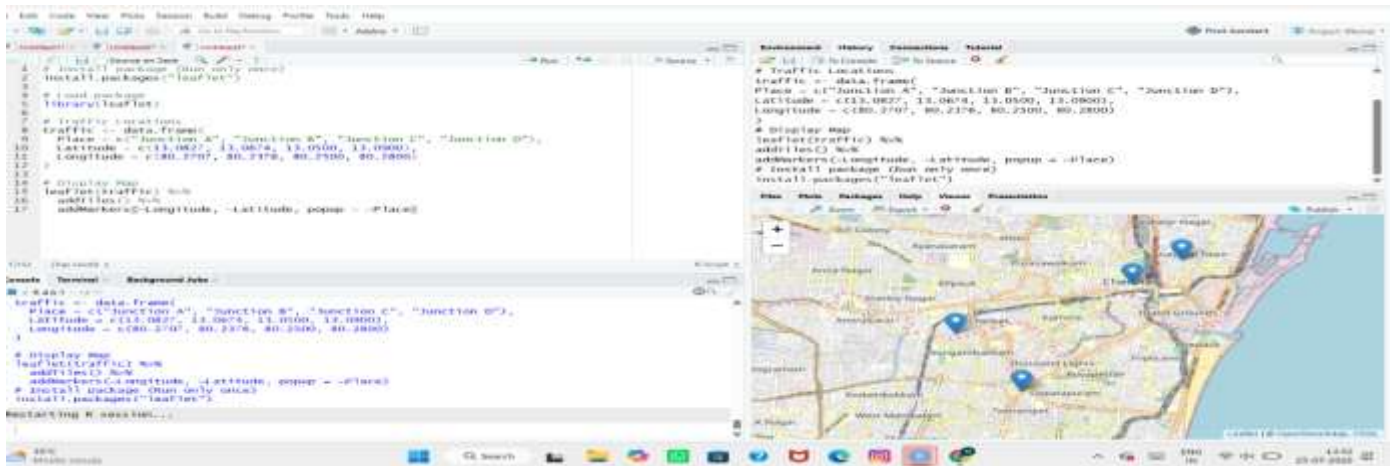
Task: Propose visualization methods including geospatial maps, heatmaps, temporal plots, and multivariate visualizations. Describe how these visualizations can assist in traffic management, route optimization, and infrastructure planning.

Introduction

A smart city collects traffic data such as vehicle counts, traffic congestion levels, average speeds, and travel times from different roads and locations. Data visualization helps city planners analyze traffic patterns, identify congestion hotspots, evaluate transportation efficiency, and make better decisions for traffic management, route optimization, and future infrastructure development.

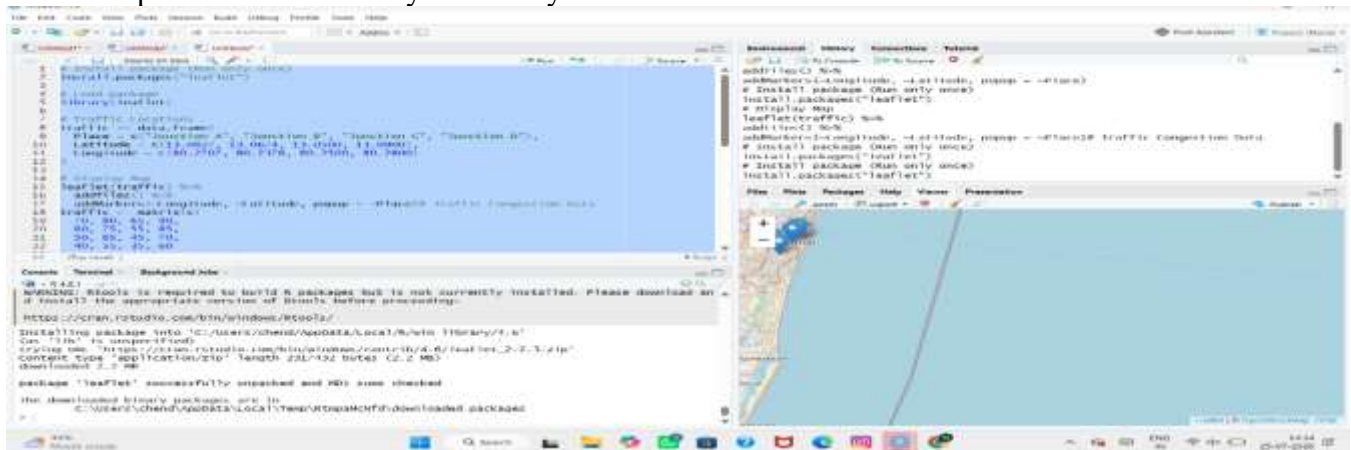
1. Geospatial Map

One-line explanation: A geospatial map displays traffic data on a city map, helping identify congestion hotspots and high-traffic locations.



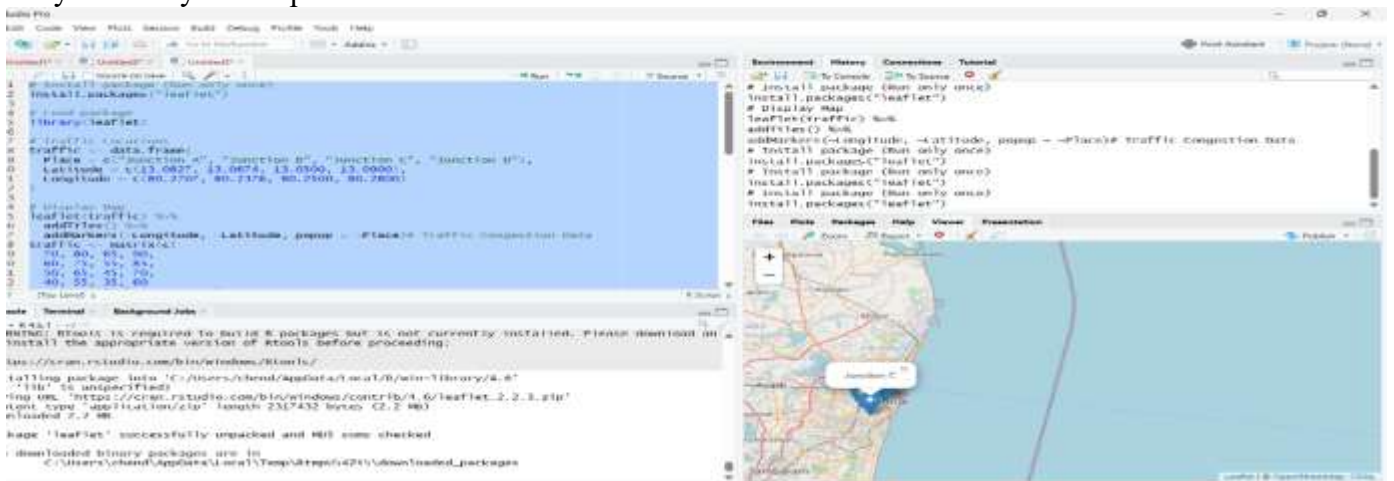
2. Heatmap

One-line explanation: A heatmap uses color intensity to represent traffic congestion levels, making busy roads and peak traffic areas easy to identify.



3. Temporal Plot (Line Chart)

One-line explanation: A temporal plot shows how traffic conditions change over time, helping analyze daily or hourly traffic patterns.



4. Multivariate Visualization (Scatter Plot/Bubble Chart)

One-line explanation: A multivariate visualization compares multiple traffic variables, such as vehicle count, speed, and travel time, to understand their relationships.



Conclusion

Geospatial maps, heatmaps, temporal plots, and multivariate visualizations provide valuable insights into traffic conditions. They help identify congestion hotspots, monitor traffic trends, optimize routes, and support efficient infrastructure planning. By using these visualization techniques, smart cities can improve transportation systems, reduce traffic congestion, and provide safer and more efficient travel for citizens.

4.A digital marketing company tracks user engagement metrics such as likes, shares, comments, viewing duration, and follower growth across multiple campaigns. The goal is to analyze relationships between engagement variables and campaign success.

Task: Suggest suitable visualization techniques such as scatter plots, heatmaps, network visualizations, and multivariate plots. Explain how these visualizations help identify influential factors, audience behavior patterns, and effective marketing strategies.

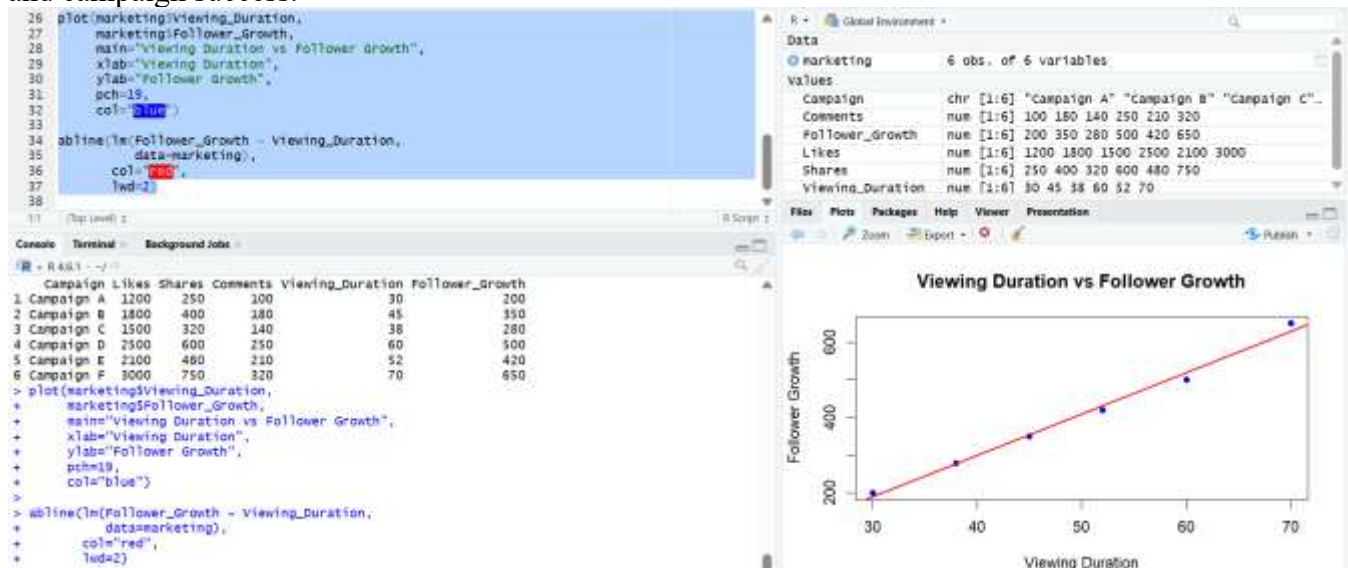
Aim: A digital marketing company can use scatter plots, heatmaps, network visualizations, and multivariate plots to analyse relationships between likes, shares, comments, viewing duration, follower growth, and overall campaign success.

1. Scatter Plot

A scatter plot can show the relationship between two engagement variables.

- X-axis → Viewing Duration
- Y-axis → Follower Growth
- Each point → One campaign

Use: Helps determine whether campaigns with longer viewing duration result in greater follower growth and campaign success.

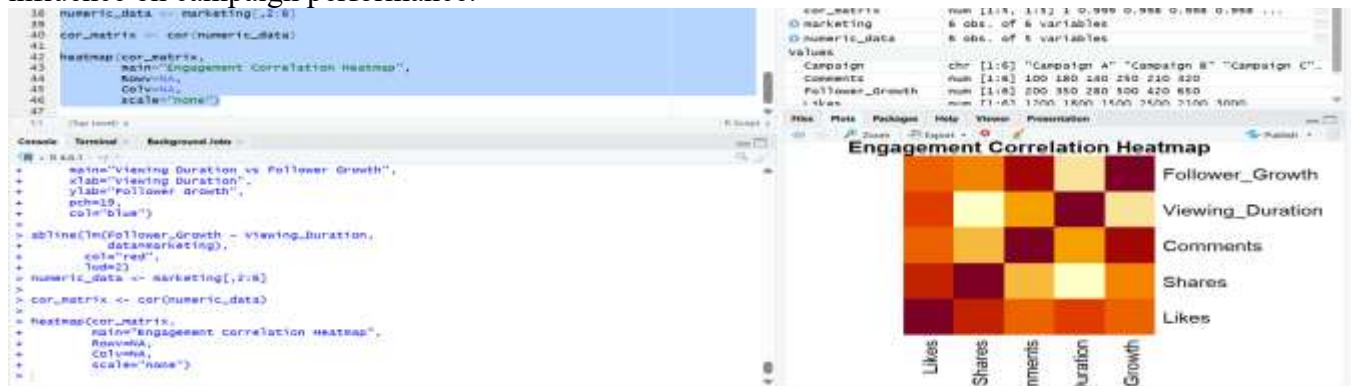


2. Heatmap

A correlation heatmap can display relationships among multiple engagement metrics.

- Variables → Likes, Shares, Comments, Viewing Duration, Follower Growth
- Color intensity → Strength of correlation

Use: Helps identify which engagement variables are strongly related and which factors have the greatest influence on campaign performance.



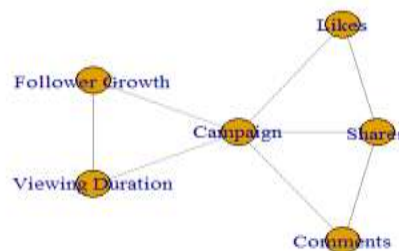
3. Network Visualization

A network visualization can represent relationships between campaigns and engagement activities.

- Nodes → Campaigns and engagement metrics
- Connections → Relationships between them

Use: Helps identify highly connected engagement factors and understand how different audience interactions contribute to campaign success.

Marketing Engagement Network



4. Multivariate Plot

A bubble plot can represent several engagement variables simultaneously.

- X-axis → Likes
- Y-axis → Shares
- Bubble size → Comments
- Color → Campaign success or follower growth

Use: Helps compare campaigns and identify combinations of engagement metrics associated with successful campaigns.

Conclusion

These visualizations help the company identify influential engagement factors, audience behavior patterns, successful campaigns, and relationships among engagement metrics. The findings can support better content creation, audience targeting, campaign optimization, and marketing strategies.

5.A healthcare organization maintains records of patients' ages, disease categories, treatment durations, recovery rates, and hospital visits. Medical administrators want to compare recovery distributions among patient groups and identify healthcare trends. Task: Select appropriate visualizations such as ridgeline plots, density plots, violin plots, histograms, and boxplots. Explain how these visualizations support disease analysis, treatment evaluation, and healthcare resource planning.

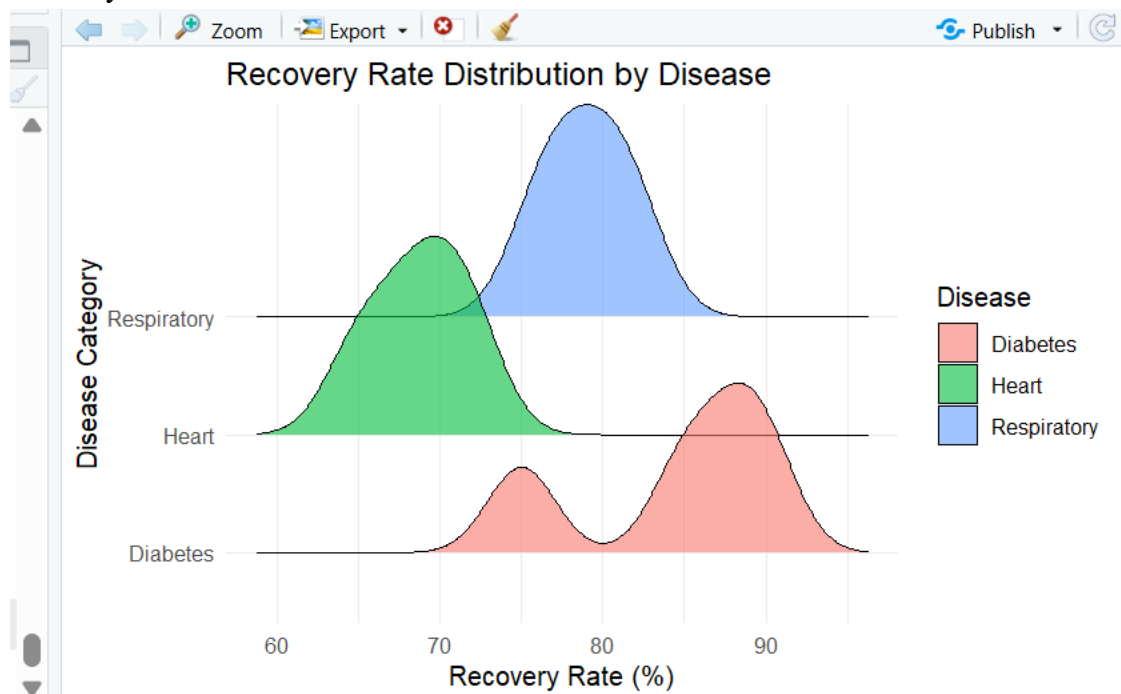
Aim: A healthcare organization can use **ridgeline plots, density plots, violin plots, histograms, and boxplots** to analyse patient age, disease categories, treatment duration, recovery rates, and hospital visits.

1. Ridgeline Plot

A **ridgeline plot** can compare recovery-rate distributions across different disease categories.

- X-axis → Recovery Rate
- Y-axis → Disease Category
- Each curve → Distribution for one disease group

Use: Helps compare recovery patterns across diseases and identify patient groups with better or poorer recovery outcomes.

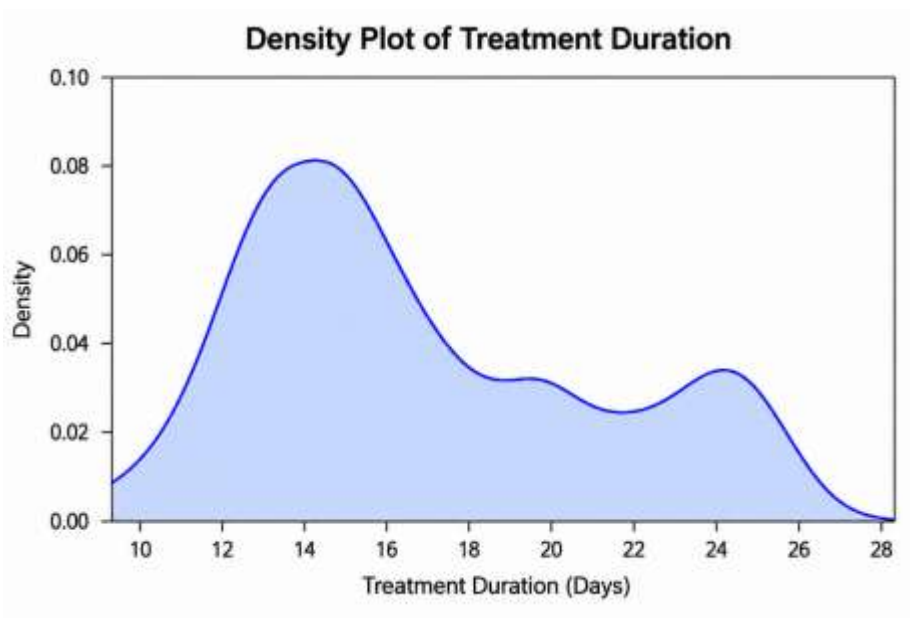


2. Density Plot

A **density plot** can show the distribution of treatment duration.

- X-axis → Treatment Duration
- Y-axis → Density

Use: Helps determine the most common treatment durations and identify whether some patients require unusually long treatment.

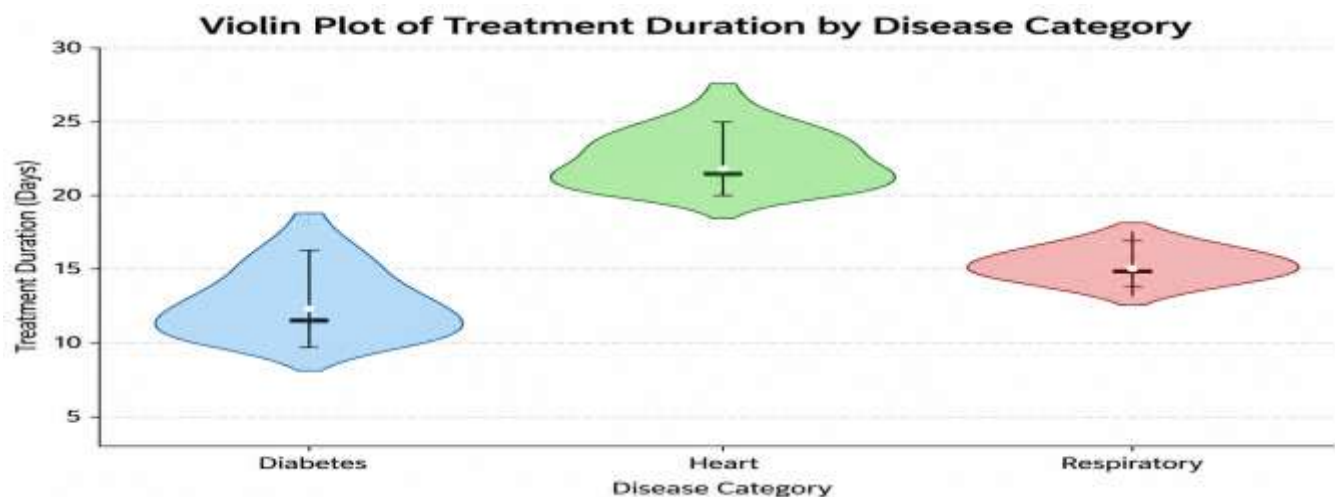


3. Violin Plot

A **violin plot** can compare treatment duration across disease categories.

- X-axis → Disease Category
- Y-axis → Treatment Duration
- Width → Concentration of patients

Use: Helps compare treatment duration, variability, and concentration among different disease groups.

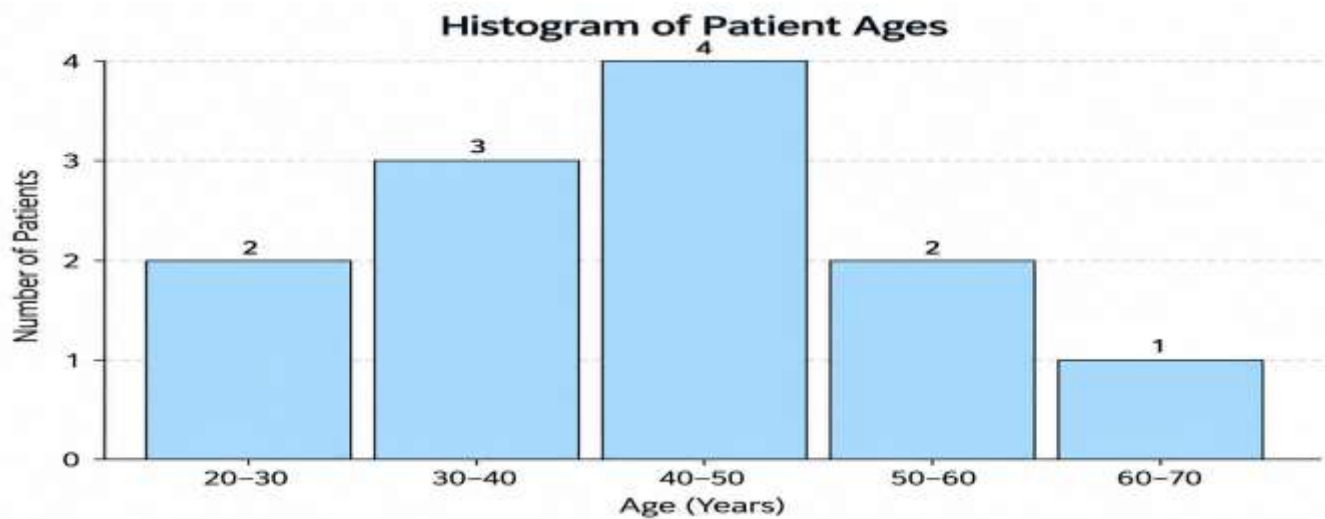


4. Histogram

A **histogram** can represent the distribution of patient ages.

- X-axis → Age groups
- Y-axis → Number of Patients

Use: Helps identify the most common patient age groups and supports age-specific healthcare planning.



Conclusion

These visualizations help healthcare administrators **compare disease patterns, evaluate treatment effectiveness, identify variations in patient recovery, and understand patient demographics**. The findings can support better allocation of **hospital beds, medical staff, treatment facilities, and other healthcare resources**.

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks	
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context		

Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts		
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution		
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools		
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____